RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 14/01/2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all the questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the three main streams that constitute a Supply Chain Management flow.	1	1	CO1
2.	State how an EIS system provides a basic operational prerequisite for global supply chain operations.	1	2	CO1
3.	Give two examples of common cybersecurity threats specific to an EIS environment.	2	2	CO4
4.	Define Agile Project Management based on its delivery characteristics.	2	2	CO4
5.	List two primary operational advantages a business gains by adopting a Cloud-based EIS (SaaS model).	2	2	CO4
	Part – B			
1	Define the terms “Web-Enabled ERP” and the “Extended Enterprise.” Explain why the transition to web technology was essential for maximizing the value of ERP in a supply chain context.	10	2	CO1
2	Clearly state the primary scope and focus of a standalone Enterprise Resource Planning (ERP) system versus a dedicated Supply Chain Management (SCM) system.	10	2	CO2
3	Define Artificial Intelligence (AI) from an Enterprise Information Systems (EIS) perspective. List and briefly explain four distinct business functions within an EIS (e.g., SCM, CRM) where AI is being successfully deployed.	10	1	CO4
4	What is the Internet of Things (IoT)? And explain its foundational role in modernizing Enterprise Resource Planning (ERP) systems. Detail how IoT enhances real-time visibility in production and supply chain monitoring.	10	1	CO4
5	Discuss the concept of “Cloud-Native” EIS architecture. Explain why this approach, utilizing micro-services and APIs, is considered more flexible and scalable than traditional monolithic cloud migration methods.	10	2	CO4

Academic year 2024-2025(Odd Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**10+50**
Course Code	**IM255TBB**	**Duration**	**120 Min**
Sem	**V Semester**	**Improvement CIE**	
ENTERPRISE INFORMATION SYSTEMS			

Note:

1. Answer all the Questions.

PART – A

Sl. No	**Questions**	**M**	**BT**	**CO**
1	Define Business Process Automation (BPA).	02	L2	CO3
2	Name two common tools used in Robotic Process Automation (RPA).	02	L2	CO3
3	Mention one advantage of implementing automation in business processes.	02	L1	CO2
4	What is a key characteristic of structured data?	02	L1	CO1
5	What are the three types of data analytics?	02	L2	CO2

PART – B

Sl. No	**Questions**	**M**	**BT**	**CO**
1.	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA). Discuss how implementing RPA can streamline the process and improve efficiency.	10	L3	CO2
2.	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenario.	10	L3	CO3
3.	Design a simplified approach to integrating Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes.	10	L4	CO4
4.	A retail chain plans to improve its inventory management system using data warehousing. Explain how a data warehouse can help in managing inventory more effectively. Include key components of a data warehouse and their roles.	10	L3	CO3
5.	Explain how data analytics in Enterprise Information Systems (EIS) can be used to improve the decision-making process for a manufacturing company aiming to reduce operational costs.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz Marks	02	04	04	–	04	06	–	–	–	–
	Test Marks	–	10	20	20	–	10	30	10	–	–

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**V**	**CIE – Improvement**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	List two functional roles of inventory.	02	1	1
2.	Identify two key factors in inventory problem analysis.	02	2	3
3.	How does an EOQ model without shortages work?	02	2	2
4.	How does inventory contribute to production efficiency?	02	2	2
5.	How is Markov Analysis used in business decision-making?	02	2	3
	Part B			
1.	The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that the cost of placing an order is Rs 36 and the cost of carrying inventory is 25 per cent of the investment in the inventories. The price is Rs 10 per kg. Help the purchase manager to determine an ordering policy for raw material.	10	1	1
2.	Each unit of an item costs a company Rs 40. Annual holding costs are 18 per cent of unit cost of the item due to miscellaneous changes: 1 per cent for insurance, 2 per cent allowances for obsolescence, Rs 2 for building overheads, Rs 1.50 for damage and loss, and Rs 4 miscellaneous costs. The annual demand for the item is constant at 1,000 units. Placing each order costs, the company Rs 100. i. Calculate EOQ and the total costs associated with stocking the item. ii. If the supplier of the item will only deliver batches of 250 units, how are the stock holding costs affected? iii. If the supplier relaxes his order size requirement, but the company has limited warehouse space and can stock a maximum of 100 units at any time, what would be the optimal ordering policy and associated costs?	10	2	2
3.	A product is sold at the rate of 50 pieces per day and is manufactured at a rate of 250 pieces per day. The set-up cost of the machines is Rs 2,000 and the storage cost is found to be Re 0.15 per piece per day. With labour charges of Rs 3.20 per piece, material cost at Rs 2.10 per piece and overhead cost of Rs 4.10 per piece, find the minimum cost batch size if the interest charges are 8 per cent (assume 300 working days in a year). Compute the optimal number of cycles required in a year for manufacturing of this product.	10	3	3

4.	A commodity is to be supplied at a constant rate of 200 units per day. Supplies of any amount can be obtained at any required time, but each ordering costs Rs 50; cost of holding the commodity in inventory is Rs 2 per unit per day while the delay in the supply of the item induces a penalty of Rs 10 per unit per day. Find the optimal policy (Q, t), where t is the reorder cycle period and Q is the inventory after reorder. What would be the best policy to adopt if the penalty cost becomes infinite?	10	3	3
5.	List and explain the characteristics of Markov Chain	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	02	04	04	–	02	08	–	–	–	–
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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Date : 18th June 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning And Design		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List any two characteristics of fixed automation systems.	(02)	1	1
2.	What is Flexible Manufacturing System (FMS)?	(02)	2	2
3.	State the concept of Just-In-Time (JIT) manufacturing.	(02)	3	3
4.	Why is a Flexible Manufacturing System (FMS) more suitable than fixed automation for product variety?	(02)	2	4
5.	Why is fixed automation not suitable for low-volume customized products? Give two reasons	(02)	2	3
	Part – B			
1.	Discuss characteristics and applications in manufacturing industries of fixed automation systems.	(10)	2	2
2.	Describe the working of a Flexible Manufacturing System (FMS).	(10)	3	2
3.	Explain the operation of a single-stage multi-machine system with a neat diagram.	(10)	3	3
4.	Illustrate how Just-In-Time (JIT) manufacturing contributes to waste elimination and productivity improvement. Explain its key principles.	(10)	3	4
5.	Propose a comprehensive plan to transform a traditional manufacturing facility into a lean production system based on JIT concepts.	(10)	2	4

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Date : 17-06-2026	**CIE – III**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is 'mass customisation' and which two principles underpin it in services?	2	1	2
2.	What is the 'market-operations gap' and what risk does it pose for service organisations?	2	2	3
3.	When a bank introduces a mobile app for all routine transfers, moving the labor from the teller to the user, what role is the customer now playing?	2	2	2
4.	What is meant by customer participation in services?	2	2	4
5.	A fast-food chain uses average service time per customer as its only performance measure. What is the risk?	2	2	3
	Part – B			
1	Discuss the Hackman and Oldham Job Characteristics Model in the context of motivating service providers. How do specific job dimensions lead to better psychological states?	10	2	4
2.	Explain the different types of service processes and their importance in service operations management.	10	2	4
3.	Prestige Home Services (PHS) is a premium home maintenance company offering plumbing, electrical, carpentry, and deep-cleaning services in Hyderabad. Technicians are dispatched to customer homes and are expected to assess the problem, carry out repairs, and collect payment - all independently. PHS recently won a large corporate contract requiring 200 service visits per month across office premises. To meet this demand, PHS hired 40 new technicians rapidly with minimal on boarding. Within 3 months, customer complaints surged. Investigations revealed that technicians were arriving without knowing the scope of the job, improvising repairs without proper tools, quoting inconsistent prices to customers, and in several cases behaving rudely when customers questioned their work. Experienced technicians reported feeling demoralised as their reputation was being damaged by new hires. The operations manager is tasked with redesigning both the service process and the people management system before the corporate contract renewal in 4 months. (a) Using service blueprinting, redesign the end-to-end service process for PHS, identifying customer actions, front-stage actions, back-stage actions, support processes, and key fail points with solutions.	10	3	3

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Date : 18th JUNE 2026	**CIE Improvement**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Instructions to Students:
1. Answer all questions
2. Use of Statistical tables and Orthogonal Arrays tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary purpose of a "Linear Graph" in Taguchi's orthogonal array designs?	2	2	CO3
2.	What is the "Strategy for Constructing an Orthogonal Array" when the number of factors is high but the budget is low.	2	3	CO3
3.	Illustrate an application of the Dummy Level Technique for a factor with 2 levels in an L_9 (3-level) array.	2	2	CO3
4.	What is the primary purpose of a "Verification Experiment" in the robust design cycle?	2	2	CO2
5.	Contrast "Testing Conditions" with "Control Factor Levels" in a typical study.	2	2	CO2
	Part – B			
1	In an investigation on the stability of power factor of capacitor it was decided to examine the following factors: **Sl No** / **Factor** / **Levels** 1 / Leakage Current of Foil (A) / $\leq$ 2 mA / $\geq$ 2 mA 2 / Power Factor of Foil (B) / $\leq$ 3% / $\geq$ 3% 3 / Capacitance Value of Coil © / 8-10 μf / 11-13 μf 4 / Soaking Time during Impregnation / 2 hrs / 4 hrs 5 / Loading Pattern / 70 % / 100 % Suspected interaction effects are AB, BC. Develop the Orthogonal Array Layout and experimental plan.	15	4	CO3

<table>
<tr><td>2</td><td>In a study to examine process control parameter for joining “BUS-BAR” of an electrical system the following factors are levels were identified for investigation with respect to efficiency of joining
<table>
<tr><th rowspan="2">Factor Code</th><th rowspan="2">Factor Description</th><th colspan="2">Level</th></tr>
<tr><th>1</th><th>2</th></tr>
<tr><td>A</td><td>Torque</td><td>25 ft lb</td><td>40 ft lb</td></tr>
<tr><td>B</td><td>Compound</td><td>Nil</td><td>Silicon Grease</td></tr>
<tr><td>C</td><td>Surface Finish</td><td>As is</td><td>Ground</td></tr>
<tr><td>D</td><td>Joint Overlap</td><td>8 mm</td><td>16 mm</td></tr>
</table>
Develop an orthogonal array design to estimate all the main effects and interaction effects A X B and C X D.</td><td>15</td><td>4</td><td>CO3</td></tr>
<tr><td>3</td><td>Provide a concise summary of major Shainin Techniques.</td><td>10</td><td>2</td><td>CO2</td></tr>
<tr><td>4</td><td>Enumerate the four types of Signal-to-Ratios for the static problems</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

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Date : 25th June 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
IM365TDA - FACILITIES PLANNING and DESIGN		

Note:
1. Answer all the Questions.

PART A

Q No.1	Questions	M	BT	CO
1.1	State one trend in facilities planning aligned with Industry 4.0.	02	2	CO1
1.2	What does a pull system in JIT manufacturing imply?	02	1	CO1
1.3	Why is reducing Work-In-Process (WIP) important in manufacturing?	02	2	CO1
1.4	List the storage operations.	02	1	CO4
1.5	Write two principles of order picking.	02	2	CO4

Part –B

Sl. No	Questions	M	BT	CO
1	Explain the concept of Flexible Manufacturing Systems (FMS). How do they improve production efficiency compared to fixed automation?	10	L2	CO1
2	Differentiate between single-stage multi-machine systems and multi-stage systems. Give examples of each.	10	L3	CO3
3	Define fixed automation systems. What are their advantages and limitations in mass production environments?	10	L2	CO2
4	Sketch and explain the functions in the warehouse.	10	L3	CO4
5	Enumerate and elaborate shipping principles.	10	L1	CO4